Examiner only

4. (a) A group of students were investigating the temperature changes during four reactions, **A**, **B**, **C** and **D**.

Two chemicals were mixed together for each reaction and the temperature change calculated.

The results of their experiment are shown in the table.

Reaction	Temperature before mixing (°C)	Temperature after mixing (°C)	Temperature change (°C)
A	21	31	+10
B	19	37	+18
C	19	....	−4
D	21	16	−5

(i) **Complete the table** by giving the temperature after mixing for reaction **C**. [1]

(ii) Give the letters, **A**, **B**, **C** or **D**, of the **two exothermic** reactions. Give the reason for your choice. [1]

Reactions .... and ....

Reason ....

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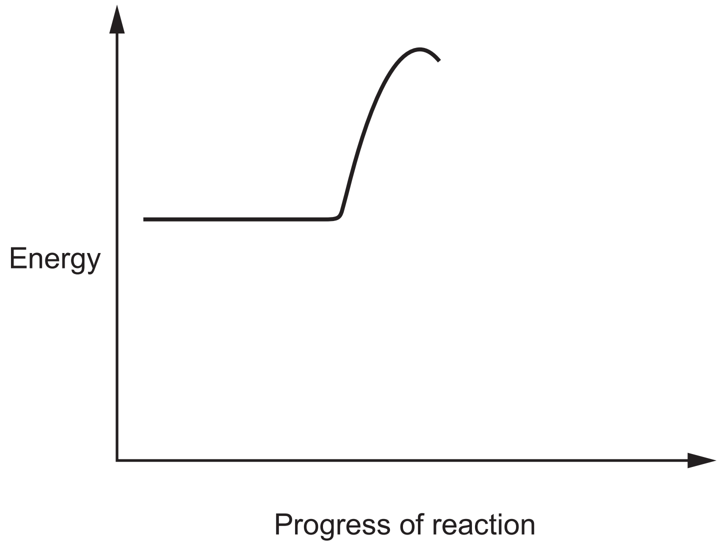
(b) Complete the energy profile diagram to show an exothermic reaction. [1]



(c) In experiment **B**, aluminium powder was mixed with zinc chloride solution. The reaction produced zinc metal and aluminium chloride solution.

Complete the symbol equation for the reaction by [2]

- writing the formula of aluminium chloride on the dotted line
- putting a number in the box to balance the equation

$$2\,Al + \boxed{}\,ZnCl_2 \longrightarrow 2 \,\ldots\ldots\ldots\ldots + 3\,Zn$$

5

